

Ke Qiu

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[[Homepage](#)] [[Google Scholar](#)]

EDUCATION

PhD Candidate | Zhejiang University

September 2022 – July 2027 (expected)

Control Science and Engineering (Robotics). Supervisor: Profs. [Yue Wang](#), [Haojian Lu](#), [Rong Xiong](#)

BEng (Chu Kochen Honors Program) | Zhejiang University

September 2018 – July 2022

Mechanical Engineering. Minor in Mathematics. GPA 3.96/4.0 (89.84/100), Top 5%

RESEARCH EXPERIENCE

External force estimation in continuum robots: Closed-form solution, solver and validation [[Video](#)]

- Derived a closed-form solution for external force estimation of Kirchhoff rods, and extended it to multisectional tendon-driven continuum robots. Developed a high-order state estimation algorithm with Gaussian process regression, deriving prior mean and kernel functions from stochastic differential equations and fusing discrete noisy measurements into posterior state distributions. Exploited Gauss–Newton algorithm for solving sparse block tridiagonal systems with $O(n)$ time complexity. Achieved superior estimation accuracy and noise robustness with a 10x speedup in computational efficiency, compared to conventional non-linear solvers (e.g., BFGS). Real-time applications are demonstrated.

Optimal path tracking control in actuator space for tendon-driven continuum robots [[Paper](#)] [[Code](#)] [[Media](#)]

- Based on the inverse kinematics, applied dynamic programming to find the shortest actuator path. Derived optimal time allocation considering actuator constraints. Designed and implemented a control framework, where trajectories planned offline serve as feedforward and inverse Jacobian computed online as real-time feedback, achieving accurate tip pose tracking. Built a multisectional tendon-driven continuum robot hardware and developed the corresponding software (C++) and interface (PLC) to achieve >100 Hz stable asynchronous real-time closed-loop control.

Efficient multi-solution solver for the inverse kinematics of 3-section constant-curvature robots [[Paper](#)] [[Page](#)]

- Formulated an analytical reduction theory that compresses a highly non-linear system of equations into a one-dimensional problem. Implemented a bounded error approximation to traverse the feasible space with resolution completeness. Developed a solver in MATLAB that eliminates initial values and local minima dependency of traditional non-linear optimisers, achieving 85% reduction in computational latency while providing multiple solutions.

Riemannian optimisation for kinematic closed-chain robots

- Modelled kinematic closed-chain robots with joint constraints as bounded Riemannian submanifolds. Developed numerical tangent-space projection and retraction methods for state updates subject to high-dimensional non-linear equality constraints, and tangent-cone projection for joint-limit equality constraints. Developed a motion generation algorithm for highly redundant closed-chain robots, enabling shape control with improved efficiency and stability.

Diffusion-based motion planning for continuum robots

- Formulated motion planning as a path generation problem in the configuration space based on the kinematics model. Trained a diffusion model to learn the inverse mapping from end-effector positions to robot configurations, generating multiple feasible solutions. Trained a diffusion model conditioned on the start and goal configurations to generate smooth intermediate sequences, enabling point-to-point and obstacle-avoidance motion planning in the workspace.

Certiably global optimal sequential state estimation

- Relaxed the robot state estimation problem in the time domain (or the arc length domain for continuum robots) into a semidefinite programming (SDP) problem using the Cayley map, minimising discrete observation errors subject to boundary constraints. Introduced redundant quadratic constraints to maintain the tightness of relaxation. Solved the resulting SDP using CVX + MOSEK in MATLAB, and verified tightness by the rank of solution matrix.

INTERNSHIP EXPERIENCE

Intern | Zhejiang Humanoid Robot Innovation Center Co. Ltd., Ningbo, China

- Dynamic parameter identification and control optimisation

November 2024 – January 2025

Developed a regression algorithm to identify latent system parameters, including joint friction and damping coefficient. Designed robust data collection and statistical fitting pipelines. Analyse the frequency spectra of reinforcement learning control signals via FFT, according to which the control bandwidths are optimised and the optimal ranges of control parameters are tested experimentally.

- Inverse kinematics analysis of parallel joint mechanisms *July 2024 – August 2024*
Modelled non-linear constrained kinematics for parallel-link ankle and waist joint mechanisms. Developed a numerical solving algorithm and implemented a C++ toolkit that achieves stable computation at 500 Hz. Analysed statics of both serial and parallel mechanisms, solving for the boundaries of permissible workspace and analysing torque transmission for motion planning and motor selection.

Intern | ABB Corporate Research Center, ABB Engineering (Shanghai) Ltd., Shanghai, China

- High-precision on-site calibration for industrial manipulators *July 2021 – August 2021*
Applied Lie group $SE(3)$ and Lie algebra $se(3)$ to build geometric models for 6DOF industrial manipulators. Proposed an on-site calibration procedure and algorithm. Constructed a non-linear least-squares problem using canonical blocks, and calibrated latent model parameters with the Levenberg–Marquardt algorithm, reducing absolute positioning errors.

PUBLICATIONS

External force estimation in continuum robots: Closed-form solution, solver and validation

Under Review. The International Journal of Robotics Research (**IJRR**), 2026

Ke Qiu, Fei Wang, Sven Lilge, Yujie Cui, Rong Xiong, Haojian Lu, Yue Wang

An actuator space optimal kinematic path tracking framework for tendon-driven continuum robots

The International Journal of Robotics Research (**IJRR**), 2025

Ke Qiu, Hongye Zhang, Jingyu Zhang, Rong Xiong, Haojian Lu, Yue Wang

An efficient multi-solution solver for the inverse kinematics of 3-section constant-curvature robots

Robotics: Science and Systems (**RSS**), 2023

Ke Qiu, Jingyu Zhang, Danying Sun, Rong Xiong, Haojian Lu, Yue Wang

Learning-based dynamics modeling and robust control for tendon-driven continuum robots

International Conference on Intelligent Robots and Systems (**IROS**), 2026

Ziqing Zou*, Ke Qiu*, Fei Wang, Haojian Lu, Rong Xiong, Yue Wang

Learning the inverse kinematics of magnetic continuum robot for teleoperated navigation

International Conference on Intelligent Robots and Systems (**IROS**), 2024

Pingyu Xiang*, Ke Qiu*, Danying Sun, ..., Yue Wang, Rong Xiong, Haojian Lu

Reinforcement learning of serpentine locomotion for a snake robot

International Conference on Real-time Computing and Robotics (**RCAR**), 2021

Ke Qiu, Hang Zhang, Yikai Lv, Yunkai Wang, Chunlin Zhou, Rong Xiong

AWARDS

IROS Best Paper Award Finalists	2025
A4X Robotics Scholarship	2025
Outstanding Postgraduate Student, Zhejiang University	2025, 2023
Outstanding Graduate of Zhejiang Province, Zhejiang Provincial Government	2022
National Scholarship, Ministry of Education, China	2021
First-Class Scholarship, Zhejiang University	2021, 2020, 2019
First-Class Scholarship, Chu Kochen Honors College	2021, 2020
First-Class Scholarship, Department of Mechanical Engineering	2020, 2019

ACADEMIC SERVICE

Reviewer for: IEEE Transactions on Robotics (**T-RO**), IEEE/ASME Transactions on Mechatronics (**TMECH**), IEEE Transactions on Automation Sciences and Engineering (**T-ASE**), Cyborg and Bionic Systems (**CBS**, Science Partner Journal), IEEE International Conference on Robotics and Automation (**ICRA**), IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**).

SELF-ASSESSMENT

Strong mathematical foundation with end-to-end research and development experience in robotics. Experienced in non-linear optimisation, state estimation, motion planning, and robot control, with a strong ability to tackle unexpected and novel technical challenges. Adaptable and fast to learn new concepts and technologies, with strong communication and collaboration skills. Responsible, proactive, and highly motivated.

邱轲

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教育经历

博士(在读) | 浙江大学

2022 年 9 月 – 2027 年 7 月 (预计)

控制科学与工程(机器人方向). 导师: [王越](#), [陆豪健](#), [熊蓉](#)

学士(竺可桢荣誉项目) | 浙江大学

2018 年 9 月 – 2022 年 7 月

机械工程. 辅修数学. 均绩 3.96/4.0, 均分 89.84/100, 排名前 5%

研究经历

连续体机器人全身外力感知: 静力学方程闭式解与高阶状态估计器 [\[Video\]](#)

提出弹性杆力学模型的外力估计闭式解理论, 并推广到多段线驱动连续体机器人. 设计基于高斯过程回归的高阶状态估计算法, 从随机微分方程导出先验均值与核函数, 融合离散测量获得状态后验分布. 回归采用 Gauss-Newton 法求解稀疏分块三对角线性系统, 时间复杂度 $O(n)$. 对比实验表明该算法具有更高力估计精度和噪声鲁棒性, 计算效率较传统数值优化(如 BFGS)提升约一个数量级. 有效应用于动态力感知与临床场景.

连续体机器人驱动空间最优路径跟踪控制 [\[Paper\]](#) [\[Code\]](#) [\[Media\]](#)

在逆运动学解的基础上, 使用动态规划计算驱动空间最短距离路径, 考虑关节约束分配最短时间. 搭建包括离线最优轨迹前馈和在线逆雅可比反馈的控制框架, 实现末端位姿高精度跟踪. 设计搭建多段线驱动连续体机械臂硬件, 编写上位机软件(C++)与底层接口(PLC), 实现高实时异步通信和 >100 Hz 稳定闭环控制.

常曲率模型逆运动学的高效多解求解器 [\[Paper\]](#) [\[Page\]](#)

针对三段等曲率建模的连续体机器人非线性逆运动学方程, 推导解析条件将原方程等价简化为一维问题, 引入有界误差近似从而遍历求解, 保障分辨率完备性. 开发 MATLAB 求解器, 相比传统数值求解器无需提供初值, 改善成功率, 总体计算效率提升 85%, 同时支持多解输出.

基于黎曼优化的闭运动链机器人运动控制

将运动学闭环机器人及其关节约束建模为带边界的黎曼子流形. 用数值方法构造切空间投影和拉回映射实现高维非线性等式约束下的状态更新, 构造切锥投影处理关节限位不等式约束. 提出高冗余度闭链机器人的运动生成算法, 实现闭链形状控制, 提升求解速度与稳定性.

基于扩散模型(Diffusion)的连续体机器人路径规划

基于分段常曲率运动学模型, 将连续体机器人末端运动规划表述为六维构型空间中的路径生成问题. 训练扩散模型学习末端位置到构型的一对多映射, 引入正运动学梯度引导, 生成多个可行逆运动学解. 进一步训练以起终点构型的扩散模型, 生成平滑的中间构型序列, 实现三维工作空间中的点到点及避障规划.

可验证全局最优性的机器人时序状态估计

使用 Cayley 映射将机器人在时域上位姿变换(或连续体机器人在弧长域上)过程中的状态估计问题松弛为半正定规划问题, 最小化离散观测误差且满足边界条件约束. 提出冗余二次约束保持紧性. 在 MATLAB 中使用 CVX + MOSEK 求解, 以解矩阵的秩一性检验松弛紧性, 实现具有可验证全局最优性的时序状态估计.

实习经历

实习生 | 浙江人形机器人创新中心有限公司, 宁波

2024 年 11 月 – 2025 年 1 月

动力学参数辨识与控制参数优化: 实现关节摩擦力与阻尼系数的参数辨识算法, 设计实验进行数据采集与拟合计算. 对强化学习策略输出的控制信号进行频域分析(FFT), 根据主要控制频率优化底层驱动器带宽, 通过实验确定最佳运动控制参数范围.

并联关节机构的逆运动学分析: 建立踝关节与腰关节所使用的并联机构的非线性带约束运动学模型, 推导数值求解算法, 开发 C++ 工具包实现 500 Hz 稳定解算. 分析串并联连杆机构静力学, 求解容许工作空间边界及力矩耦合关系, 为强化学习训练提供可行域参数.

实习生 | ABB中国研究院, ABB工程(上海)有限公司, 上海

2021 年 7 月 - 2021 年 8 月

- 工业机械臂现场高精度标定算法: 基于李群 $SE(3)$ 与李代数 $se(3)$ 建模六自由度工业机械臂正向运动学, 提出一套工业现场标定流程及算法, 使用标准工件提供的末端位置约束构造非线性最小二乘问题, 使用Levenberg-Marquardt算法迭代求解各关节真实转轴参数, 实现工业现场快速标定, 补偿模型误差提升绝对定位精度.

发表论文

在IJRR, RSS等机器人领域重要国际期刊和会议上发表论文 5 篇:

External force estimation in continuum robots: Closed-form solution, solver and validation
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Robotics: Science and Systems (RSS), 2023
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Learning-based dynamics modeling and robust control for tendon-driven continuum robots
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Learning the inverse kinematics of magnetic continuum robot for teleoperated navigation
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Reinforcement learning of serpentine locomotion for a snake robot
International Conference on Real-time Computing and Robotics (RCAR), 2021
Ke Qiu, Hang Zhang, Yikai Lv, Yunkai Wang, Chunlin Zhou, Rong Xiong

发明专利

授权发明专利 2 项:

- 基于高斯过程回归的连续体机器人状态估计方法 CN122432468A
一种细长弹性杆的外力估计方法 CN122413777A

荣誉奖励

IROS最佳论文提名 2025	积加机器人奖学金 2025
浙江大学优秀研究生 2025, 2023	浙江省优秀毕业生 2022
国家奖学金 2021	竺可桢学院创新奖学金 2021, 领航奖学金 2020
浙江大学一等奖学金 2021, 2020, 2019	机械工程学院国际精密奖学金 2020, 汉商液压奖学金 2019

学术兼职

作为以下期刊及会议审稿人: IEEE Transactions on Robotics (T-RO), IEEE/ASME Transactions on Mechatronics (TMECH), IEEE Transactions on Automation Sciences and Engineering (T-ASE), Cyborg and Bionic Systems (CBS, Science Partner Journal), IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

自我评价

具备扎实的数理基础与机器人全栈研发能力. 在非线性优化, 状态估计, 运动规划与控制等方面均有深入研究. 善于学习新知识, 乐于沟通协作. 有责任心, 进取心和工作热情.